

2-40 图 2-188 所示三铰拱截面 K 的弯矩为 _____ (下拉为正), 拱轴线方程为 $\frac{4f}{l^2}(lx-x^2)$ 。(广西大学 2010)

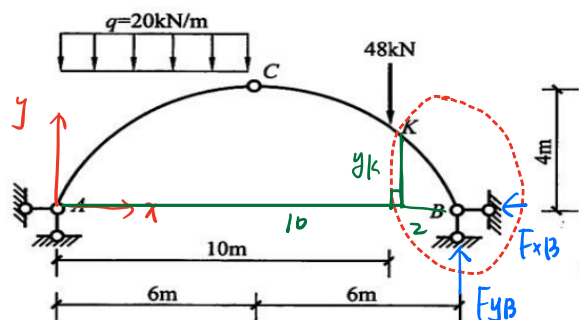


图 2-188

$$\sum M_A = 0, 12F_{yB} - 48 \times 10 - 20 \times 6 \times 3 = 0, \therefore F_{yB} = 70 \text{ kN}$$

$$\sum M_C = 0, 48 \times 4 + 4F_{xB} - 6F_{yB} = 0, \therefore F_{xB} = 57 \text{ kN}$$

$$y = \frac{4 \times 4}{12^2} (12x - x^2) = \frac{4}{3}x - \frac{x^2}{9}$$

$$\therefore y_K = \frac{4}{3} \times 10 - \frac{10^2}{9} = \frac{20}{9} \text{ m}$$

$$\begin{aligned} \sum M_K &= 0 \\ M_K + 57 \times \frac{20}{9} - 70 \times 2 &= 0 \\ \therefore M_K &= \frac{40}{3} \text{ kN}\cdot\text{m} \end{aligned}$$

2-41 图 2-189 所示三铰拱的水平支座反力是多少? (武汉大学 2010)

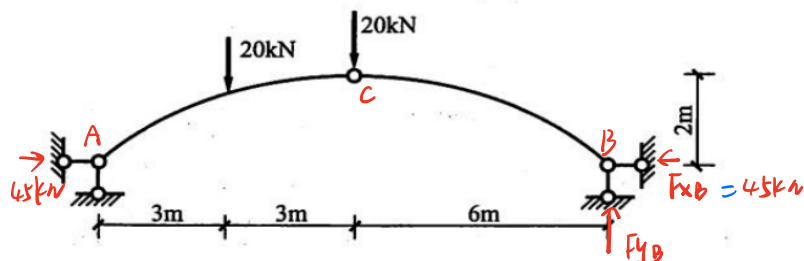


图 2-189

$$\sum M_A = 0, 12F_{yB} - 20 \times 3 - 20 \times 6 = 0, F_{yB} = 15 \text{ kN}$$

$$\sum M_C = 0, 2F_{xB} - 6F_{yB} = 0, F_{xB} = 45 \text{ kN}$$



2-42 图 2-190 所示结构中, 链杆 AB 的轴力为 $-F_P$ 。(以受拉为正)(中南大学 2010)

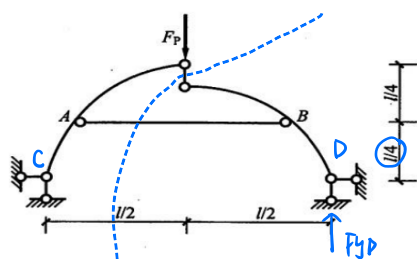


图 2-190

$$\begin{aligned}\sum M_C = 0, \\ F_{yD} \times l - F_P \times \frac{l}{2} = 0 \\ \therefore F_{yD} = \frac{F_P}{2}\end{aligned}$$

$$\begin{aligned}\sum M_D = 0 \\ F_{NAB} \times \frac{l}{4} + \frac{F_P}{2} \times \frac{l}{2} = 0 \\ \therefore F_{NAB} = -F_P\end{aligned}$$

2-43 图 2-191 所示三铰拱, 拱轴线方程为 $y = \frac{4f}{l^2} (lx - x^2)$, 其中 $f = 4m$, $l = 16m$, 则截面 D 弯矩 M_D 等于 $4q$ (外侧受拉), 截面 D 剪力 F_{QD} 等于 0。(浙江大学 2012)

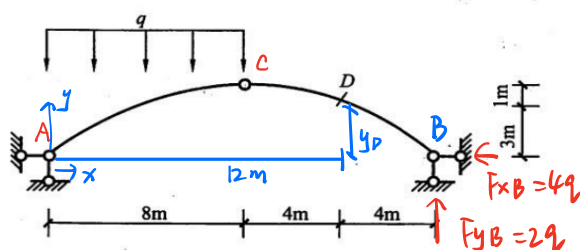


图 2-191

$$y_D = \frac{4 \times 4}{16^2} \times (16 \times 12 - 12^2) = 3$$

$$y = \frac{4 \times 4}{16^2} (16x - x^2) = x - \frac{x^2}{16}$$

$$y' = 1 - \frac{2x}{16} = 1 - \frac{x}{8}$$

$$y'_D = 1 - \frac{12}{8} = -\frac{1}{2}$$



$$\sin \theta = \frac{1}{\sqrt{5}}$$

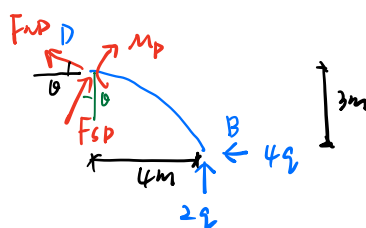
$$\cos \theta = \frac{2}{\sqrt{5}}$$

$$\sum M_A = 0,$$

$$16 F_{yB} - q \times 8 \times 4 = 0, \therefore F_{yB} = 2q$$

$$\sum M_C = 0$$

$$4 F_{xB} - 2q \times 8 = 0, \therefore F_{xB} = 4q$$



$$\sum M_D = 0, M_D + 4q \times 3 - 2q \times 4 = 0$$

$$\therefore M_D = -4q$$

$$\sum F_x = 0, F_{ND} \times \frac{2}{\sqrt{5}} + 4q - F_{SD} \times \frac{1}{\sqrt{5}} = 0$$

$$\sum F_y = 0, F_{ND} \times \frac{1}{\sqrt{5}} + F_{SD} \times \frac{2}{\sqrt{5}} + 2q = 0$$

$$4q - F_{SD} \times \frac{1}{\sqrt{5}} - \frac{2}{\sqrt{5}} F_{ND} - 2q = 0 \Rightarrow F_{SD} = 0$$

